Real-Time Step Counting and Heading Estimation on the SensorTile

Group 3: Ahil Hussain - ahus071, Inesh Bhanuka - ibha186
CS704 Project 2, University of Auckland

***Abstract*—AICorp tasked us with developing a real-time wearable pedestrian dead reckoning (PDR) prototype on the STMicroelectronics SensorTile. It monitors the location and safety of factory staff and streams data to a smartphone over Bluetooth. Satellite positioning fails indoors, so the device must rely on its own sensors. However, steel in a factory distorts the magnetic field used for heading, and staff can wear the device in any orientation. We count steps with an adaptive peak-and-valley detector on the acceleration magnitude, which does not depend on how the board is worn. Heading comes from a tilt-compensated, self-calibrating magnetometer fused with the gyroscope. The gyroscope carries the heading alone wherever the field is disturbed. The walking direction is learnt from the wearer's walking motion or set with a hand gesture. With the board in a trouser pocket, the step count was within 1.3% over 239 steps. Held flat, the heading averaged 4.3° from a phone compass. On walked routes, turns were measured within 10°, even where the magnetometer was rejected for most of each leg.**

***Index Terms*—pedestrian dead reckoning, step detection, heading estimation, sensor fusion**

I. Introduction

AICorp wants to know where its employees are inside a factory, both for safety and so that help can reach them quickly after an incident. GPS signals are blocked or reflected indoors, and fitting a building with beacons is expensive, so AICorp asks for a wearable device that tracks a person's step count and orientation by itself. We have been provided with a SensorTile kit, which carries an accelerometer and magnetometer (LSM303AGR) and a gyroscope (LSM6DSM). The sensor data is shown using the ST BLE Sensor phone app.

Step count and heading on their own do not give a location. With an estimate of each step's length, however, they are enough to rebuild a walked path from a known starting point. This is known as pedestrian dead reckoning (PDR) [1], [2]. Rather than tracking position continuously, PDR moves the position forward by one step length in the current heading at every detected step. This avoids integrating the acceleration twice, which drifts within seconds with low-cost sensors unless the sensor is mounted on the foot [1]. Our device has to stay accurate in a factory, where steel structures and machinery distort the Earth's magnetic field, and it should work however the worker carries it.

II. Background

A. Step Detection

Most step detectors look for the periodic peaks of walking in the acceleration. Jin et al. [1] rotate the acceleration into a world frame and look for a peak followed by a valley on the vertical axis. Tian et al. [2] first work out how the phone is carried (held, swinging or in a pocket) and then apply a threshold to the axis that suits that mode. Lee et al. [3] avoid both by using the magnitude of the acceleration, which does not change with orientation, with thresholds that adapt to the recent signal. They reported 99.5% accuracy over seven placements. Brajdic and Harle [4] compared eight step counters on 130 recordings. Simple windowed peak detection did best, and they recommend checking the signal's standard deviation so that no steps are counted while the user stands still. Step length is often estimated with the Weinberg model [5], which relates it to the fourth root of the acceleration range within the step.

B. Heading Estimation

Jin et al. [1] used only an accelerometer and compass, and found that metal cabinets near the walking path were one of their main sources of heading error. Tian et al. [2] integrated the gyroscope instead and learnt the offset between the phone and the walking direction, but gyroscope drift then became their largest error. Filter-based methods combine the two. Yuan et al. [6] used a quaternion unscented Kalman filter in which the gyroscope predicts the orientation and the accelerometer and magnetometer correct it, reaching under 10° mean error at the waist. Kuang et al. [7] only used the magnetometer while the field was steady, with 2.3° RMS error in the hand but 16.6° in a pocket.

TABLE I
Approaches in the reviewed literature

Work	Step detection	Heading	Reported result
[1]	Vertical peak–valley	Compass only	73.7% less error
[2]	Carrying mode first	Gyro, learnt offset	98.9% position
[3]	Adaptive, on $\lvert\mathbf{a}\rvert$	–	99.5% of steps
[4]	Eight compared	–	$<$1.3% median
[6]	–	Kalman, gyro+mag	$<10^\circ$ mean
[7]	Peak, Weinberg	Kalman, steady mag	2.3–16.6° RMS

C. Implications for This Project

Table I summarises these works. Two findings shaped our design. First, adaptive thresholds on the acceleration magnitude count steps reliably regardless of how the device is carried [3], [4]. Second, a heading from the magnetometer alone is unreliable near steel. The common solution is to add a gyroscope and to ignore the magnetometer whenever the field appears disturbed [6], [7]. Since a factory is full of steel,

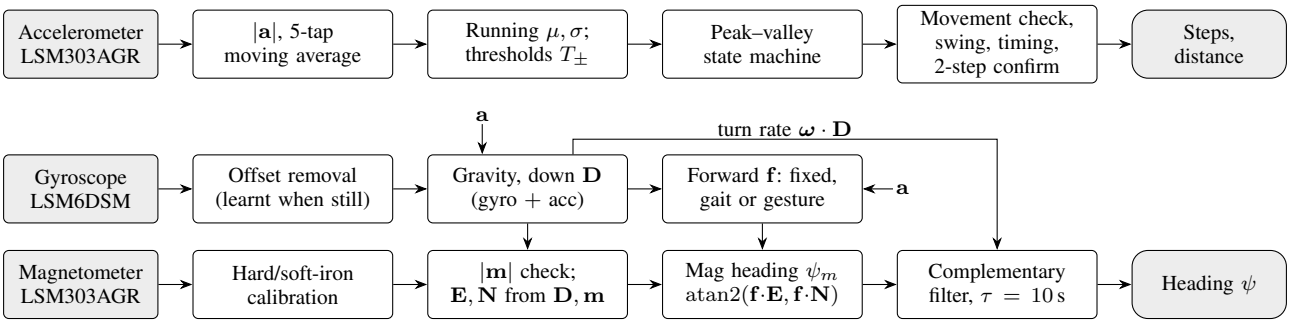


Fig. 1. Flowchart of the firmware, run once per 20 ms sample. Top: step detection. Bottom: heading estimation, where the accelerometer $\mathbf{a}$ also feeds the gravity estimate and the walking-direction learner.

we used the SensorTile's gyroscope as well to ensure heading accuracy.

III. System Design

A. Overview

The firmware runs on the SensorTile's STM32L476. Every 20 ms the main loop reads all three sensors over SPI, converts the readings to physical units and passes them to two independent modules, one for steps and one for heading (Fig. 1). The results are sent over BLE to the ST BLE Sensor app. The accelerometer and magnetometer fields carry the raw data, and the three gyroscope fields carry the step count, heading and distance.

The accelerometer runs at 50 Hz (± 4 g, 12-bit), the magnetometer at 100 Hz and the gyroscope at 52 Hz (± 500 dps). The two sensor chips are mounted at different orientations, so the firmware maps all readings into one right-handed frame.

B. Step Detection

Wherever the board is worn, each step produces a rise in acceleration above 1 g at heel strike and a dip below 1 g as the body falls into the next step. The detector counts one step for each rise followed by a dip in the magnitude $|\mathbf{a}|$, which does not depend on how the board is worn, smoothed with a 5-sample (100 ms) moving average.

The thresholds adapt to the walker. A running mean μ and standard deviation σ of the smoothed signal are kept with exponential moving averages over about one second, and the upper and lower thresholds are

$$T_{\pm} = \mu \pm \max(0.3\,\sigma,\ 35\,\text{mg}). \tag{1}$$

The 35 mg floor stops the thresholds from sinking into the sensor noise when the signal is quiet. A small state machine then records the highest point above T_+ and the lowest point after it below T_-. A peak with no valley within 1 s is thrown away.

A completed peak and valley only count as a step if three conditions hold. The standard deviation must be above 40 mg, which shows that the person is moving [4]. The swing from peak to valley must be at least 120 mg. The step must also not come too soon after the previous one: the minimum gap is half the recent average step interval, and never less than 250 ms, which stops a double bump within one step from being counted twice. Counting only starts after two valid steps in a row, so a single knock or sitting down adds nothing, and a pause longer than 2 s ends the walk.

Each step's length is estimated with the Weinberg model [5], $L = K\sqrt[4]{a_{\max} - a_{\min}}$, using the peak-to-valley swing in m/s^2 and $K = 0.45$. The distance is the sum of the step lengths.

C. Heading Estimation

1) Gravity: Tilt correction needs to know which way is down. The accelerometer gives this when the board is still, but while walking it also measures the body's motion. Each sample, the firmware rotates its previous gravity estimate using the gyroscope and then moves it 2% of the way towards the accelerometer reading, which averages the walking motion out over about a second. The downward direction $\mathbf{D}$ is the normalised, reversed gravity vector.

2) Calibration: The board's battery and circuitry add a constant offset to the magnetometer reading (hard iron, about 350 mG on the Z axis of our board) and scale each axis slightly differently (soft iron) [8]. After power-up the user turns the board so that each face points up in turn, while the firmware records the minimum and maximum on each axis. The middle of each range gives the offset, and each axis is rescaled so that all three ranges have the same radius R. The calibration is used once every axis has swung by at least 800 mG; Auckland's field is about 550 mG, so a full turn gives about 1100 mG.

3) Tilt-compensated heading: Rather than calculating roll and pitch angles, we find East and North directly in the board's axes:

$$\mathbf{E} = \mathbf{D} \times \mathbf{m}, \qquad \mathbf{N} = \mathbf{E} \times \mathbf{D}, \qquad \psi_m = \operatorname{atan2}(\mathbf{f}\cdot\mathbf{E},\ \mathbf{f}\cdot\mathbf{N}), \tag{2}$$

where $\mathbf{m}$ is the calibrated field and $\mathbf{f}$ is the direction the wearer walks in, in board axes. This works because the Earth's field always lies in the plane containing North and Down, so $\mathbf{D}\times\mathbf{m}$ points East at any tilt.

4) Gyroscope fusion: A rotation about the downward direction is a turn of the wearer, so the gyroscope rate $\boldsymbol{\omega}$ changes the heading by $(\boldsymbol{\omega}\cdot\mathbf{D})\,\Delta t$ each sample. The gyroscope's small offset, about 3 dps on one axis of our board, is learnt whenever the board is completely still. The heading follows

the gyroscope and is pulled slowly towards the magnetometer heading:

$$\psi_k = \psi_{k-1} + (\boldsymbol{\omega} \cdot \mathbf{D})\,\Delta t + \frac{\Delta t}{\tau}\,\mathrm{wrap}(\psi_m - \psi_{k-1}), \qquad (3)$$

with $\tau = 10\,\mathrm{s}$, where wrap keeps the difference between $-180°$ and $180°$. When the measured field strength differs from R by more than 20%, as it does next to steel benches, the magnetometer term is skipped and the gyroscope carries the heading alone, similar to [7].

D. Direction of Travel

Equation (3) needs the board axis that points the way the wearer walks. By default this is $+X$, but the firmware can also find it in two other ways. When the board is at the waist or in a hand, the body rocks forwards and backwards with each step more than it sways sideways. The firmware tracks the direction of the largest horizontal acceleration, which gives the walking line, and decides which end is forward from the timing of this swing relative to the vertical bounce. After 3 s of steady walking, the learnt direction replaces $+X$. This is similar to the offset learning in [2].

A board in a trouser pocket moves with the thigh, so this does not work there. Instead the user holds the board flat with $+X$ pointing ahead for 2 s. The firmware takes $+X$ as forward, uses the gyroscope to keep track of that direction while the board is put into the pocket, and fixes it to the board once the board has settled. Whenever the forward direction changes, the heading is turned by the same angle so that it does not jump. The gesture also freezes the calibration and resets the step count, holding the board face down for 3 s starts a new calibration, and the LED flashes after each of these events.

IV. System Performance

A. Testing Method

We streamed every raw sample over UART to a laptop while walking with the board, then replayed the recordings through the same step and heading code on a PC, so settings could be tried on the same walk without walking it again. All results below come from the final code. In simulation, the heading at six tilted poses was within $0.8°$ of the true value.

Early tests were done in the Embedded Systems Lab, full of steel benches, where the field strength varied between 413 and 894 mG along a single walk. The final tests used the final firmware in a room with a strong disturbance at one end (up to 1.8 times the normal field). The reference was a phone compass set to magnetic north. All trials were done with the board in a trouser pocket.

B. Step Detection

In Fig. 2 the thresholds sit about 36 mg either side of the mean while each stride swings the signal by several hundred mg. Over the 239 walked steps in Table II the board counted 242 (1.3% high), and none while standing still.

The weakness is handling. Turning the board by hand during calibration produced up to 29 false steps, so the final firmware

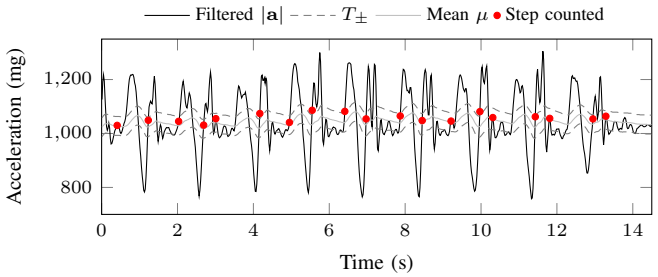


Fig. 2. One 20-step run with the board in a trouser pocket. Each step is counted when the signal climbs back above the mean after its valley; 19 of the 20 steps were counted.

TABLE II
Step count with the board in a trouser pocket

Trial	Walked	Counted
Out and back	35	35
Three straight runs	20, 20, 20	19, 19, 20
Route, legs 2–4 (7/7/15)	29	31
Reference route (15/7/7/15)	44	44
Route with turns and a U-turn	≈71	74
Total	239	242 (+1.3%)
Standing still (all trials)	0	0
Sitting down and standing up, 3 times	0	2
Pocketing after the forward gesture	0	3

resets the count at the forward gesture that ends the routine. Pocketing the board after that added 3 steps, and sitting down and standing up three times 2.

C. Heading

Fig. 3 shows the magnetometer during calibration. The raw readings are centred on $(-123, -73, 367)$ mG, mostly because of the board itself, and their strength varied from 257 to 946 mG with orientation. After calibration the readings lie close to circles centred on zero, with a strength of 551 ± 9 mG, which matches Auckland's field of about 550 mG.

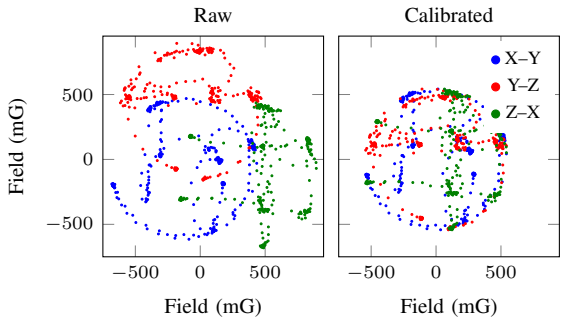


Fig. 3. Magnetometer readings while the board was turned through all six faces, before and after calibration, shown as three projections.

Table III shows the heading with the board held flat and pointed north, east, south and west using the phone compass. The mean error was $4.3°$, which is close to how accurately the board can be lined up with a phone by hand.

TABLE III
HEADING WITH THE BOARD HELD FLAT

Direction	N	E	S	W
Expected	0°	90°	180°	270°
Measured	0.2°	94.0°	187.4°	275.7°
Error	+0.2°	+4.0°	+7.4°	+5.7°

Table IV shows two walks of the same route: steps ahead (first bearing 89° and 90°), a right turn, a turn-around and a left turn. The long legs of run 1 were within about 3°. Run 2 passed the disturbed end of the room, where the magnetometer was ignored for 57–70% of every leg, so the heading came mostly from the gyroscope; it still measured the turns as 85°, 279° and 190° against 90°, 270° and 180°. Short legs are less accurate because they are mostly the turns at each end, and in a pocket the heading swings by 10–15° with every stride. Against a reference worked out from the gyroscope, the forward direction set by the gesture was within 5° at the waist and 9° in the hand. Learnt from walking, it was within 3–11° at the waist and 9–16° in the hand, and as expected it did not work in a pocket.

TABLE IV
HEADING ON TWO WALKS OF THE SAME ROUTE (POCKET)

	Run 1		Run 2	
Leg	Measured	Error	Measured	Error
1, ahead	89.3°	+0.3°	95.6°	+5.6°
2, right turn	170.9°	−8.1°	180.9°	+0.9°
3, turn-around	341.1°	−17.9°	14.3°	+14.3°
4, left turn	272.1°	+3.1°	285.5°	+15.5°
Mean absolute error		7.4°		9.1°

D. Design Choices

Several design decisions came from problems found in testing. Our first version used only the magnetometer, as suggested. Whenever the field looked disturbed, it ignored the reading and held the last good heading. The laboratory is full of steel benches that distort the field, so the heading stayed frozen for up to two and a half minutes, and any turn made during that time was missed. We therefore added the gyroscope, which measures turns without relying on the magnetic field. Replaying our lab walks with a 10 s time constant instead of 2 s halved the heading wobble, because over a few seconds in a distorted building the gyroscope is more trustworthy than the magnetometer.

A plain low-pass filter on the accelerometer gave a good gravity estimate at the waist but lagged behind a board in a pocket, which tilts by about 30° with every stride. With that lag, a 180° turn in a pocket was measured as 159°. Rotating the estimate with the gyroscope before filtering brought this to 175°.

The min/max calibration is simple, but a single magnet near the board stretches it. Two calibrations done while holding the laptop gave a radius of 616–628 mG instead of about 550 mG, so good readings failed the 20% check more often. This is why the calibration is frozen by the forward gesture and can be restarted without a reset.

Turning the board clockwise on a desk showed that left and right turns were swapped, which our earlier sensor checks could not reveal; negating the LSM303AGR's Y axis fixed it. Both modules are small, with 1.3 kB of code for steps and 4.4 kB for heading, and the loop kept its 50 Hz rate even while logging every sample.

E. Limitations

Most remaining problems come from how the board is handled. Putting it away after the forward gesture, or sitting down, can still add a few steps. These could be blocked using the gyroscope rate, which was 11–74 dps while walking but 110–150 dps while handling the board. The setup is also fragile. The calibration is lost at every reset, and the forward gesture relies on the user knowing which end is $+X$.

The heading has two further limits. The disturbance check only looks at field strength, so one laboratory spot passed with a normal strength but a dip angle of $-19°$ instead of $-54°$. Checking the dip angle as well would catch this. The heading is also relative to magnetic north, which is about 20° east of true north in Auckland. Finally, our testing was limited. The step length constant K was not calibrated, the distance was not checked against a measured course, and all trials used one person.

V. CONCLUSION

Our SensorTile prototype counts steps and reports heading in real time over Bluetooth. With the board in a pocket the step count was within 1.3% over 239 steps, and held flat the heading was 4.3° from a phone compass on average. The gyroscope kept the heading working near steel, and the forward gesture lets the board be worn any way round.

REFERENCES

[1] Y. Jin, H.-S. Toh, W.-S. Soh, and W.-C. Wong, "A robust dead-reckoning pedestrian tracking system with low cost sensors," in *Proc. IEEE PerCom*, 2011, pp. 222–230.

[2] Q. Tian, Z. Salcic, K. I.-K. Wang, and Y. Pan, "A multi-mode dead reckoning system for pedestrian tracking using smartphones," *IEEE Sensors J.*, vol. 16, no. 7, pp. 2079–2093, 2016.

[3] H.-H. Lee, S. Choi, and M.-J. Lee, "Step detection robust against the dynamics of smartphones," *Sensors*, vol. 15, no. 10, pp. 27230–27250, 2015.

[4] A. Brajdic and R. Harle, "Walk detection and step counting on unconstrained smartphones," in *Proc. ACM UbiComp*, 2013, pp. 225–234.

[5] H. Weinberg, "Using the ADXL202 in pedometer and personal navigation applications," Analog Devices, Application Note AN-602, 2002.

[6] X. Yuan, S. Yu, S. Zhang, G. Wang, and S. Liu, "Quaternion-based unscented Kalman filter for accurate indoor heading estimation using wearable multi-sensor system," *Sensors*, vol. 15, no. 5, pp. 10872–10890, 2015.

[7] J. Kuang, X. Niu, and X. Chen, "Robust pedestrian dead reckoning based on MEMS-IMU for smartphones," *Sensors*, vol. 18, no. 5, Art. no. 1391, 2018.

[8] STMicroelectronics, "AN4825: Ultra-compact high-performance eCompass module based on the LSM303AGR," Application Note.